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Inventor(s)

Good; Joshua Perry et al.

Camper Pod

Abstract

A camper pod includes a camper housing and an attachment portion. The camper housing has a housing body. The camper housing also has a pair of structure layers positioned on opposite surfaces of the housing body. The housing body is formed of a foam material. The pair of structure layers are each formed of a resin material deposited on the foam material. The attachment portion removably attaches the camper housing to a transportation element.

Inventors: Good; Joshua Perry (Lititz, PA), Janney; Adam (Lebanon, PA)

Applicant: Tusca Outdoors, LLC (Lebanon, PA)

Family ID: 1000008490788

Assignee: Tusca Outdoor, LLC (Lebanon, PA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of the filing date under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application No. 63/557,118, filed on Feb. 23, 2024.

FIELD OF THE INVENTION

[0002] The present invention relates to a camper pod and, more particularly, to an insulated and lightweight camper pod removably attachable to a transportation element.

BACKGROUND

[0003] Campers that are mountable in truck beds or otherwise attachable to vehicles are typically heavy and cumbersome, even for small campers, making them difficult for users to manipulate and attach. These campers, due to the material and construction of the camper, are also very expensive. The combination of weight and expense makes many campers unusable or inaccessible to users who desire an insulated and transportable camping structure.

SUMMARY

[0004] A camper pod includes a camper housing and an attachment portion. The camper housing has a housing body and a pair of structure layers positioned on opposite surfaces of the housing body. The housing body is formed of a foam material. The pair of structure layers are each formed of a resin material deposited on the foam material. The attachment portion removably attaches the camper housing to a transportation element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The invention will now be described by way of example with reference to the accompanying figures, of which:

[0006] FIG. 1 is a perspective view of a camper pod according to an exemplary embodiment;

[0007] FIG. 2 is a perspective view of a camper pod according to another exemplary embodiment;

[0008] FIG. 3 is a rear schematic view of the camper pod of FIG. 1 removably attached to a transportation element and having a shield;

[0009] FIG. 4 is a front perspective view of the camper pod of FIG. 1 with an attachment position removably attaching the camper pod to the transportation element;

[0010] FIG. 5A is a perspective sectional view of a portion of the camper pod of FIG. 1 and a portion of the transportation element of FIGS. 3 and 4 taken along line B-B' of FIG. 3;

[0011] FIG. 5B is another perspective sectional view of the portion of the camper pod of FIG. 5A and the portion of the transportation element of FIG. 5A taken along line B-B' of FIG. 3;

[0012] FIG. 6 is a schematic sectional view of a material of the camper pod of FIG. 1 taken along line A-A' of FIG. 1;

[0013] FIG. 7 is a rear perspective view of the camper pod of FIG. 2 having an attachment portion removably attaching the camper pod to another transportation element;

[0014] FIG. 8 is a rear perspective view of the camper pod of FIG. 1 having another attachment portion removably attaching the camper pod to the transportation element;

[0015] FIG. 9 is an enlarged perspective view of portion C of FIG. 8;

[0016] FIG. 10A is a perspective view of one accessory element of a plurality of accessory elements attached to the camper pod of FIG. 1 or the camper pod of FIG. 2;

[0017] FIG. 10B is a perspective view of two different accessory elements of the plurality of accessory elements attached to the camper pod of FIG. 1 or the camper pod of FIG. 2;

[0018] FIG. 10C is a perspective view of another accessory element of the plurality of accessory elements attached to the camper pod of FIG. 1 or the camper pod of FIG. 2;

[0019] FIG. 10D is a plan view of an attachment assembly attachable to the camper pod of FIG. 1

or the camper pod of FIG. 2; and

[0020] FIG. 11 is a flowchart of a method of assembling a camper pod according to an exemplary embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENT(S)

[0021] Exemplary embodiments of the present invention will be described hereinafter in detail with reference to the attached drawings, wherein like reference numerals refer to like elements. The present invention may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided so that the present disclosure will convey the concept of the invention to those skilled in the art.

Moreover, the embodiments and features thereof are only possible configurations in which, however, the individual features as described herein can be provided independently of one another or can be omitted.

[0022] Throughout the specification, various terms are used to describe a camper pod 1. The term “removably attachable”, or any iteration thereto, is intended to mean that an element is attachable to another element such that a secure connection between the elements is made, but the connection can easily be undone without damaging either of the elements.

[0023] A relative thickness, length, or width of the elements described in more detail below appear in FIGS. 1-10D. The relative thickness, length, or width of the elements are merely illustrative and are not meant to be representative of a particular thickness, length, or width of any of the elements.

[0024] A camper pod 1 according to an exemplary embodiment is shown in FIGS. 1, 3, and 4, and 8, and an alternative exemplary embodiment of the camper pod 1 is shown in FIGS. 2 and 7. The camper pod 1, as described below, will primarily be described with reference to the camper pod 1 according to FIGS. 1, 3, 4, and 8. Only differences between the configuration of the camper pod 1 according to FIGS. 2 and 7 and the camper pod 1 according to FIGS. 1, 3, 4, and 8 will be described in detail when referencing the alternative embodiment; a primary difference between the camper pod 1 according to FIGS. 1, 3, 4, and 8 and the camper pod 1 according to FIGS. 2 and 7 are differences in dimensions, particularly lateral dimensions.

[0025] The camper pod 1 comprises a camper housing 10, as shown in FIG. 1, and an attachment portion 50 disposed on the camper housing 10, as shown in FIGS. 4, 5A-5B, and 7-9. The attachment portion 50 is used to removably attach the camper housing 10 to a transportation element 100, as described in detail below.

[0026] The camper housing 10, as shown in FIGS. 1 and 10B, has a first roof 21 and a second roof 23 adjacent to the first roof 21 and extending from the first roof 21. The camper housing 10, as shown in FIG. 1, also has a bottom 24 opposite the first roof 21 and parallel to the first roof 21.

[0027] As shown in the embodiment of FIG. 1, the first roof 21 has an approximately flat shape and the second roof 23 has an angled shape. The second roof 23 may extend at an angle greater than 0° and less than 90° with respect to a plane of the first roof 21. In the shown embodiment, the second roof 23 extends at an angle of approximately 30° with respect to the plane of the first roof 21. In other embodiments, as shown in FIG. 2, the second roof 23 may have a less angled shape, extending at a shallower angle with respect to the plane of the first roof 21. In other embodiments, the second roof 23 may extend at any angle from the first roof 21 that is suitable for reducing wind drag on the camper pod 1. In further embodiments, the camper housing 10 only has the flat first roof 21 or the angled second roof 23.

[0028] As shown in FIG. 1, the camper housing 10 also has a pair of sidewalls 25 opposite each other, a rear wall 27, and a front wall 29 opposite the rear wall 27. The sidewalls 25, the rear wall 27, and the front wall 29 connect the bottom 24 and the roofs 21, 23. The camper housing 10, including the bottom 24, the roofs 21, 23, and the walls 25, 27, 29, defines an interior to provide a shelter for a user.

[0029] The parts of the camper housing 10, including the bottom 24, the roofs 21, 23, and the walls 25, 27, 29, are all formed of a same plurality of layers of material 11, 12, 13. The layers of the parts

of the camper housing **10**, as shown in FIGS. **5A**, **5B**, and **6**, include a pair of structure layers **11** on opposite sides of a housing body **12**, and a UV protection layer **13** on at least one of the structure layers **11**.

[0030] FIGS. **5A** and **5B** show perspective sectional views of a portion of the camper pod **1** taken along line B-B' of FIG. **3**. The portion of the camper pod **1** shown in FIGS. **5A** and **5B** is shown separated from the rest of the camper pod **1** only for clarity in illustrating the layers **11**, **12**, **13** of material of the camper housing **10** and one possible form of attachment between the camper pod **1** and a substrate **103** of a transportation element **100**; the portion of the camper pod **1** shown in FIGS. **5A** and **5B** is not separate from the camper pod **1** during use. FIG. **6** shows a schematic sectional view of the camper pod **1** taken along line A-A' of FIG. **1**, also showing the layers of material **11**, **12**, **13** of each part of the camper housing **10**.

[0031] The housing body **12**, shown in FIGS. **5A**, **5B**, and **6**, is formed of a foam material. In an embodiment, the foam material is a high-density EPS foam with a density of 2.4lbs/ft.^{sup.3}. In another embodiment, the foam material may be an EPS foam with a density greater or less than 2.4lbs/ft.^{sup.3}. In various other embodiments, the foam material may be any type of closed-cell foam that allows the housing body **20** to function as described herein. In another embodiment, the foam material is a composite foam material formed of a foam adhered with an aluminum layer. The aluminum layer may be a thin gauge aluminum layer.

[0032] The pair of structure layers **11**, shown in FIGS. **5A**, **5B**, and **6**, are disposed on opposite sides of the housing body **12**. The structure layers **11**, as described below, are coated on the sides of the housing body **12** in a manner that is not removable from the housing body **12**. The structure layers **11** are each formed of a resin material. In an embodiment, the resin material is a pure aromatic polyurea, such as a 100% pure aromatic polyurea. In various other embodiments, the resin material may be a hybrid polyurea, such as a polyurea mixed with a polyurethane, an aliphatic polyurea, or any other material that allows the structure layers **11** to function as described herein.

[0033] As shown in FIGS. **5A**, **5B**, and **6**, the UV protection layer **13** is disposed on a side of at least one of the structure layers **11** opposite the housing body **12**. In an embodiment, the UV protection layer **13** is disposed on both structure layers **11** on the side opposite the housing body **12**. The UV protection layer **13**, in an embodiment, is a layer of UV paint that is adhered to the structure layer **11**. In another embodiment, the UV protection layer **13** may be a UV protective material that is integrated into the resin material of the structure layers **11**.

[0034] In the embodiment shown in FIGS. **1** and **4**, the camper housing **10** also comprises an embedded strip **15**. The embedded strip **15** is positioned on the first roof **21** and extends between the side walls **25**. In an embodiment, the embedded strip **15** has a pair of protrusions that extend from the first roof **21** and form a groove **17** between them. In the shown embodiment, the camper housing **10** has a pair of embedded strips **15** that are attached to the first roof **21**, parallel to one another, and spaced apart from one another, with one positioned adjacent to the rear wall **17** and one positioned adjacent to the second roof **23**. In an embodiment, the embedded strip **15** is made of a composite wood material. In various other embodiments, the embedded strip **15** may be made of any other structurally stable material.

[0035] A method of assembling the camper pod **1** shown in FIGS. **1**, **3**, **4**, and **8**, and the alternative embodiment of the camper pod **1** shown in FIGS. **2** and **7**, is now described primarily with reference to FIG. **11**.

[0036] The different embodiments of the camper pod **1**, particularly shown in FIGS. **1** and **2**, are similarly assembled. The camper pods **1** according to the various embodiments, however, have different lengths and/or widths (e.g., different sized panels of the foam material that form the housing body **12**). While assembly of the camper pod **1** is described in a specific order herein, the sequence of the steps are merely exemplary; the sequence of the steps may be changed, additional steps may be added, and/or steps may be completed simultaneously. Attachments to the housing body **12** may also be interchangeably described as attachments to, or elements of, the camper

housing **10**; thus, as described herein, when an element is described as being attached to the camper housing **10**, the element is also attached to the housing body **12**, the UV protection layer **13** and/or the pair of structure layers **11**.

[0037] A plurality of foam panels made of the foam material of the housing body **12** are cut in various sizes and shapes in a first step **200**. The foam panels are cut to form various parts **21**, **23**, **24**, **25**, **27**, **29** of the camper pod **1** and overall sizes of the camper pod **1**. Two exemplary embodiments of the camper pod **1** shown in FIG. **1** may have the following dimensions and weights once assembled: (1) outside dimensions: 79" L×58" W×54" H; interior dimensions: 74.5" L×54" W×48" H; weight: 180 lbs; and (2) outside dimensions: 89" L×58" W×54" H; interior dimensions: 84.5" L×54" W×48" H; weight: 190 lbs. An exemplary embodiment of the camper pod **1** shown in FIG. **2** may have the following dimensions and weight once assembled: outside dimensions: 89" L×48" W×54" H; interior dimensions: 84.5" L×44" W×48" H; weight: 188 lbs or 180-190 lbs. In other embodiments, the camper pod **1** may have other exterior dimensions and other interior dimensions in order to adapt the size of the camper pod **1** to various transportation elements **100**. The camper pod **1** may also be lighter or heavier than in the shown embodiments, for example any weight less than 200 lbs., but in all embodiments is sufficiently light to be manually liftable by a group of people.

[0038] In an embodiment, each of the foam panels of the housing body **12** has a thickness of at least 2". In an embodiment, the thickness of the housing body **12** is different at different parts of the camper housing **10**; in an exemplary embodiment, the thickness of the foam panel is 2.5" on the first roof **21**, the second roof **23**, and the bottom **24**, and the thickness of the foam panel is 2" on the pair of sidewalls **25**, the rear wall **27**, and the front wall **29**. In various other embodiments, the thickness of the foam panels of the housing body **12** that form the parts **21**, **23**, **24**, **25**, **27**, **29** of the camper housing **10** may be the same or different on the various parts **21**, **23**, **24**, **25**, **27**, **29** of the camper housing **10** and may be greater or smaller than the thicknesses described herein.

[0039] The cut foam panels of the housing body **12** are then attached together in a second step **201**, shown in FIG. **11**. In an embodiment, the panels of the housing body **12** are attached using an adhesive, such as an industrial hot glue, and fasteners, such as screws. First, the adhesive is applied to the cut foam panels. The foam panels are then assembled to form the structure of the camper housing **10** having a shape as shown in FIG. **1** or **2**. Second, before the adhesive cures, the fasteners are screwed into the foam panels to secure the foam panels in position.

[0040] Any rough edges of the assembled panels of the housing body **12** are then sanded flat in a third step **202**. A joint compound is used to cover any seams between adjacent cut foam panels, or holes created by the fasteners, in a fourth step **203**. Another sanding step of sanding the joint compound may occur after the joint compound dries for between 6-8 hours.

[0041] The embedded strip **15**, or a plurality of embedded strips **15**, as shown in FIGS. **1-2**, are then attached to the first roof **21** in a fifth step **204**. In an alternative embodiment, the embedded strip(s) **15** can be attached anywhere on the assembled housing body **12** panels, or alternatively, may be omitted. In various embodiments, the embedded strip(s) **15** are attached to the first roof **21** with an adhesive, a fastener, or in any way that secures the embedded strip(s) **15** in the position described herein.

[0042] Interior corners of an interior of the assembled housing body **12** are then caulked in a sixth step **205**. In an embodiment, all interior corners are caulked and then the caulk is allowed to dry.

[0043] The resin material is then deposited on the foam panels of the housing body **12** to form the pair of structure layers **11** in a seventh step **206** shown in FIG. **11**. During the seventh step **206**, a proportioner is used to spray the resin material on the interior and the exterior of the housing body **12** on opposite surfaces of the housing body **12** to form an interior structure layer **11** and an exterior structure layer **11**. The proportioner is also used to spray the resin material on the embedded strip **15**. In an alternative embodiment, the embedded strip **15** is not attached on the first roof **21** of the housing body **20** until after the seventh step **206**.

[0044] In the seventh step **206**, the proportioner sprays the resin material as follows: (1) the resin material and an isocyanate (ISO) are separately heated, (2) the heated resin material and the heated ISO are pumped through individual hoses to a tip of a spray gun of the proportioner, (3) the heated resin material and the heated ISO are combined at the tip of the spray gun, and (4) the combined heated resin and the heated ISO are sprayed onto the foam material of the housing body **12** and the embedded strip(s) **15** at a high pressure, such as a pressure of approximately 3200 psi, and a high temperature, such as a temperature greater than 150 degrees Fahrenheit. In an embodiment, the proportioner is a two-part proportioner. In various other embodiments, the aforementioned pressure and temperature may be any pressure and temperature that adheres the resin material combined with the ISO on the housing body **12** to form the structure layers **11** on the housing body **12**. In an exemplary embodiment, a thickness of the structure layers **11** deposited on the housing body **12** is greater than or equal to 2/32" and less than or equal to 5/32".

[0045] The UV protection layer **13** is then applied onto the structure layers **11**, such as by painting, in an eighth step **207**. In an embodiment, the UV protection layer **13** is UV protective paint that is painted onto an outer surface of the exterior structure layer **11** or both of the structure layers **11**. The eighth step **207** may be omitted if the resin material used in the seventh step **206** is the aliphatic polyurea or another type of resin material that includes UV protection material; in this embodiment, the UV protection layer **13** and the structure layers **11** are the same layers.

[0046] The camper housing **10** of the camper pod **1** is assembled at the end of step **207**.

[0047] Other elements of the camper pod **1**, such as a door **40**, a port **42**, a vent **44**, a pair of windows **46**, an angled shield **48**, and a plurality of accessory elements **60**, may then be attached to the camper housing **10**, as shown in FIGS. **1**, **3**, and **10A-10C**, in a ninth step **208** shown in FIG. **11**. The aforementioned other elements of the camper pod **1** may also be omitted or placed on any part of the camper housing **10** (e.g., the port **42** is on the rear wall **27** in the embodiment shown in FIG. **1**, but not on the rear wall **27** on the embodiment shown in FIG. **2**), or additional other elements may be attached to the camper housing **10**.

[0048] As shown in FIGS. **1-3**, the door **40** is attached to an opening in the rear wall **27**. The opening in the rear wall **27** is sized to fit the door **40** and is either formed during the first step **200** or after the second step **201** by cutting the foam panels of the housing body **12**. The door **40** is a fiberglass camper style door. In an embodiment, the door **40** may have a door window **41**, as shown in FIG. **10C**. In various other embodiments, the door **40** may be any type of door that provides access to an interior of the camper pod **1**.

[0049] As shown in FIG. **3**, the angled shield **48** may be attached to the rear wall **27** above a top of the door **40**. In various embodiments, the angled shield **48** may be attached to the rear wall **27** with an adhesive, a fastener, or in any way that provides environmental protection for the seams of the door **40**.

[0050] As shown in FIG. **1**, the port **42** is attached to an opening in the rear wall **27**. The opening in the rear wall **27** is sized to fit the port **42** and is either formed during the first step **200** or after the second step **201** by cutting the foam panels of the housing body **12**. In an exemplary embodiment, the port **42** is a 3" electrical port. In various other embodiments, the port **42** may be any type of outlet that allows the camper pod **1** to function as described herein. In various other embodiments, the camper pod **1** may have any number of outlets **42**.

[0051] As shown in FIGS. **1-2**, the vent **44** is attached to an opening in the sidewall **25**. The opening in the sidewall **25** is sized to fit the vent **44** and is either formed during the first step **200** or after the second step **201** by cutting the foam panels of the housing body **12**. In an exemplary embodiment, the vent **44** is a 3" vent. In various other embodiments, the vent **44** may be any type of vent that allows the camper pod **1** to function as described herein. In another embodiment, both sidewalls **25** have the vent **44**. In yet another embodiment, the vent **44** is a 12V powered fan vent on the sidewall(s) **25**, and/or a 12V powered fan on the first roof **21** and/or the second roof **23**.

[0052] As shown in FIGS. **1-2** and **7-8**, the pair of windows **46** are each attached to an opening in

one sidewall **25**. The opening in each sidewall **25** is sized to fit each window **46** and each opening is either formed during the first step **200** or after the second step **201** by cutting the foam panels of the housing **12**. In an exemplary embodiment, each window **46** is a standard RV window that slides and has a screen. In various other embodiments, the windows **46** may be any type of window used in campers or RVs. In another embodiment, only one window **46** is attached to one sidewall **25**. In various other embodiments, the camper pod **1** may have any number of windows **46**.

[0053] After the door **40**, the port **42**, the vent **44**, and the windows **46** are attached to the housing body **12** in the step **208**, silicone is applied around the door **40**, port **42**, vent **44**, and windows **46**.

[0054] As shown in FIGS. **10A-10C**, the plurality of accessory elements **60** attached to the camper housing **10** in the step **208** can include a solar panel **61** attached on the second roof **23**, a storage rack **63** attached on the first roof **21**, and a fan **65** attached on the rear wall **27**. In various embodiments, the solar panel **61** is attached to the second roof **23** with an adhesive, a fastener, or in any way that allows the solar panel **61** to function as described herein. In various embodiments, the storage rack **63** is attached to the first roof **21** with an adhesive, a fastener, or in any way that allows the storage rack **63** to function as described herein. The fan **65** is attached to the camper housing **10** in the same way as the vent(s) **44**. The fan **65** may be installed on the rear wall **27**, as shown in FIG. **10C**, or in lieu of any of the vent(s) **44**.

[0055] The plurality of accessory elements **60** may also include various fixtures. For example, the plurality of accessory elements **60** may include a carpet, air conditioning, heaters, heat pumps, handles, hooks, etc. The carpet, for example, may be attached to the bottom **24** on the interior of the housing body **20** with, for example, an adhesive.

[0056] The attachment portion **50** is then attached to the camper housing **10** in a tenth step **209** shown in FIG. **11**. Various configurations of the attachment portion **50** are described below; however, these configurations are merely exemplary. In various other embodiments, the attachment portion **50** may include any element, and may be attached anywhere on the camper housing **10**, that allows the attachment portion **50** to attach the camper housing **10** to the transportation element **100**. Different embodiments of the attachment portion **50** are shown in FIGS. **4**, **5A-5B**, and **7-9**. Various configurations of the attachment portion **50** are now described.

[0057] As shown in FIG. **4**, a first configuration of the attachment portion **50** includes a strap **53**. As shown in FIG. **4**, the strap **53** is received by the groove **17** of the embedded strip **15** and is fed along the groove **17** and around the sidewalls **25**. In an embodiment, the strap **53** is a ratchet strap. In various other embodiments, the strap **53** may be any type of strap that allows the strap **53** to function as described herein.

[0058] As shown in FIGS. **5A-5B**, a second configuration of the attachment portion **50** includes a plurality of plates **51** and a plurality of fasteners **55**. A hole or plurality of holes are drilled in the bottom **24** on the exterior of the housing body **20** for each plate **51**. Each plate **51** is then secured to the bottom **24** with the fasteners **55**. In an embodiment, as shown in FIG. **5B**, each plate **51** has an approximately square shape and the plurality of plates **51** comprises more than two of the plates **51**. In another embodiment, the plurality of plates **51** only includes two of the plates **51**. The two plates **51** of another embodiment of the attachment portion **50**, as shown in FIG. **10D**, each have an approximately rectangular shape.

[0059] In an embodiment, the fasteners **55** may include washers, bolts, and locking nuts. In an exemplary embodiment, the washers may be 4"×4" 14-gauge steel washers and 5/16 washers, the bolts may be 5/16-18×4.5" stainless steel bolts, and the locking nuts may be 5/16-18 nylon locking nuts. In various other embodiments, the fasteners **55** may be any type of fastener that allows the attachment portion **50** to attach the camper housing **10** to the transportation element **100**.

[0060] As shown in FIGS. **7-9**, a third configuration of the attachment portion **50** includes a plurality of plates **51**, the fasteners **55**, and a tie down ring **59**. Holes are first drilled into the camper housing **10**. For example, holes are drilled into the sidewall **25**, or the pair of sidewalls **25**, for each plate **51**. Each plate **51** is then secured to the sidewall **25** with the fasteners **55**. The tie

down ring **59** is then secured to the plate **51** and the sidewall **25** with the fasteners **55**. In an embodiment, the tie down **57** is a turnbuckle tie down. In another embodiment, the tie down ring **59** is a D-ring.

[0061] A transportation element **100** is now described with reference to FIGS. 3-5B and 7-9. The transportation element **100** may be any type of transportation device, such as a truck with a truck bed or a trailer, that can have the camper pod **1** removably attached to it. The transportation element **100** may have a substrate **103**, such as a floor or floorboards, as shown in FIGS. 3-5B. The transportation element **100** may also have an anchor **101**. In an embodiment, the anchor **101** is a D-buckle anchor.

[0062] Removable attachment of the camper pod **1** to the transportation element **100** with the various configurations of the attachment portion **50** is now described with reference to FIGS. 3-5B and 7-9. A camper according to an embodiment of the invention includes the camper pod **1** and the transportation element **100**.

[0063] As shown in FIG. 4, the first configuration of the attachment portion **50** is removably attached to the transportation element **100** with the strap **53**. The strap **53**, or a plurality of straps **53**, are attached to the camper pod **1** as described above and then a portion of the strap **53** is fed under the transportation element **100**, as shown in FIG. 4. The camper pod **1** is then removably attached to the transportation element **100** when each strap **53** or the strap **53** is tightened and secured, for example by a ratchet. In various other embodiments, the strap **53** may be fed on/under any part of the transportation element **100** that allows the camper pod **1** to be removably attached to the transportation element **100** with the strap **53**.

[0064] As shown in FIGS. 3 and 5A-5B, the second configuration of the attachment portion **50** is removably attached to the transportation element **100** with the plurality of plates **51**, and the fasteners **55**. A hole for each plate **51** is drilled in the bottom **24** as described above and then a plurality of complementary holes, each lining up with each hole in the bottom **24**, are drilled in the substrate **103**, such as a bed of a trailer. Each plate **51** is then placed on the exterior structure layer **11** and/or the interior structure layer **11** in alignment with the holes. The fasteners **55** are then placed through the holes to attach the camper pod **1**, via the bottom **24**, to the transportation element **100**. In another embodiment, the plates **51** may be omitted. In various other embodiments, the holes drilled into the bottom **24** and the substrate **103** may be threaded and the fasteners **55** may only include a bolt and washer instead of a bolt, washer, and nut.

[0065] As shown in FIGS. 8-9, the third configuration of the attachment portion **50** is removably attached to the transportation element **100** with the tie down ring **59**, a tie down **57**, a platform **72**, and anchors **101**, with the camper pod **1** positioned in a truck bed. The platform **72** is used to raise the camper pod **1** over wheel wells of the transportation element **100**. The platform **72** may have internal storage space. First, as shown in FIG. 8, the platform **72** is placed on the transportation element **100**. The camper pod **1** is then placed on the platform **72**. Each tie down **57**, as shown in FIG. 9, is then attached between each tie down ring **59** and each anchor **101**. The camper pod **1** is then removably secured to the transportation element **100** when each tie down **57** is tightened.

[0066] As shown in FIG. 7, the third configuration of the attachment portion **50** is removably attached to the transportation element **100** with the tie down ring **59**, the tie down **57**, and optionally a spacer **70** or a plurality of spacers **70**. The spacer(s) **70** are used to fill a space between the camper pod **1** and a portion of the transportation element **100**, as shown in FIG. 7. The camper pod **1** of the embodiment shown in FIG. 7 fits between wheel wells of the transportation element **100**. First, as shown in FIG. 7, the camper pod **1** is placed on the transportation element **100** between the wheel wells. The spacers **70** are then placed against the pair of sidewalls **25** and the portion of the transportation element **100**. Each tie down **57** is then attached to each tie down ring **59** and each anchor **101** as described with respect to FIGS. 8 and 9. The camper pod **1** is then removably secured to the transportation element **100** when each tie down **57** is tightened.

[0067] The camper pod **1** removably attached to the transportation element **100** according to FIGS.

7-9 has each anchor **101** attached on a rail or in walls of a bed or a bottom of the bed of the transportation element **100**. However, each anchor **101** may be attached on any portion of the transportation element **100** that allows the camper pod **1** to be removably attached to the transportation element **100** with the third configuration of the attachment portion **50**.

[0068] The embodiments of the camper pod **1** have a lightweight construction; in various embodiments, the camper pod **1** weighs no more than 200 lbs, not including the weight of any of the plurality of accessory elements **60** that may be attached to the camper pod **1**. The camper pod **1**, unlike other typical campers, is sufficiently lightweight to be manually loaded onto the transportation element **100** with only a few people, such as two people, or even a single person. Additionally, the camper pod **1** is sufficiently durable to withstand being transported on the transportation element **100** and used in the elements. The structure layers **11** cure to a hardness that provides durability, longevity, and chemical/water resistance to the camper pod **1**. For example, the pure aromatic polyurea provides optimal durability, while a hybrid polyurea provides sufficient durability while being more cost effective.

[0069] The camper pod **1**, according to an exemplary embodiment, is sized to fit a single person or a pair of people. The camper pod **1** is also sufficiently insulated, by the material of the housing body **12** and the structure layers **11**, such that the body heat of the individual(s) inside can provide sufficient heat for multiple seasons. For example, body heat alone can keep an inside temperature of the camper pod **1** between 50-55 degrees Fahrenheit when a temperature of an environment surrounding the camper pod **1** is 20 degrees Fahrenheit. The port **42** also permits the use of an additional heater or other electrical elements if necessary.

Claims

1. A camper pod, comprising: a camper housing having a housing body and a pair of structure layers positioned on opposite surfaces of the housing body, the housing body is formed of a foam material and the pair of structure layers are each formed of a resin material deposited on the foam material; and an attachment portion removably attaching the camper housing to a transportation element.
2. The camper pod of claim 1, wherein the resin material is a pure polyurea material, a hybrid polyurea material, or an aliphatic polyurea material.
3. The camper pod of claim 1, wherein the foam is a closed cell-foam having a density of at least 2.4 lbs/ft.^{sup.3}.
4. The camper pod of claim 1, wherein the housing body has a thickness of at least 2".
5. The camper pod of claim 4, where each of the structure layers has a thickness greater than or equal to 2/32" and less than or equal to 5/32".
6. The camper pod of claim 1, wherein a UV protection layer is applied on at least one of the structure layers.
7. The camper pod of claim 1, wherein a UV protection layer is integrated into at least one of the structure layers.
8. The camper pod of claim 1, wherein the camper housing has a first roof and a second roof adjacent to the first roof, the second roof has an angle greater than 0° and less than 90° with respect to a plane of the first roof.
9. The camper pod of claim 8, wherein the camper housing has an embedded strip extending across the first roof and receiving a ratchet strap of the attachment portion.
10. The camper pod of claim 1, wherein the attachment portion has a plate disposed on a bottom of the camper housing, the plate removably attaches the bottom of the camper housing to a substrate of the transportation element.
11. The camper pod of claim 10, wherein the attachment portion includes a plurality of fasteners extending through the plate, the bottom of the camper housing, and the substrate of the

transportation element.

12. The camper pod of claim 1, wherein the attachment portion has a plate attached to a sidewall of the camper housing, a tie down ring attached to the plate, and a tie down attached between the tie down ring and an anchor of the transportation element.

13. The camper pod of claim 1, further comprising a door and an outlet attached to a rear wall of the housing body, a pair of windows each attached to one sidewall of the housing body, and a vent attached to the one sidewall of the housing body.

14. The camper pod of claim 1, further comprising at least one of: a storage rack attached on a roof of the housing body, a solar panel attached on a roof of the housing body, and a fan attached to a rear wall of the housing body.

15. The camper pod of claim 1, wherein the camper pod has a weight of less than or equal to 200 lbs.

16. A camper, comprising: a transportation element; and a camper pod including a camper housing having a housing body and a pair of structure layers positioned on opposite surfaces of the housing body, the housing body is formed of a foam material and the pair of structure layers are each formed of a resin material deposited on the foam material, the camper pod has an attachment portion removably attaching the camper housing to the transportation element.

17. The camper of claim 16, wherein the transportation element is a truck bed or a trailer.

18. A method of removably attaching a camper pod to a transportation element, comprising: providing the camper pod having a weight less than or equal to 200 lbs; manually lifting the camper pod; placing the camper pod on the transportation element; and removably attaching the camper pod to the transportation element with an attachment portion of the camper pod.

19. The method of claim 18, further comprising a plurality of steps of assembling the camper pod, including: cutting a plurality of foam panels of a housing body; attaching the plurality of foam panels of the housing body together; and spraying a resin material on an interior of the housing body and an exterior of the housing body.

20. The method of claim 19, wherein the spraying step includes separately heating the resin material and an isocyanate, combining the heated resin material and the heated isocyanate, and spraying the resin material combined with the isocyanate on the interior of the housing body and the exterior of the housing body.
